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Review

Deep learning in financial fraud detection: innovations, challenges, and applications

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ABSTRACT

This study presents a systematic literature review of 108 peer-reviewed publications (2019–2024) on the application of deep learning (DL) to financial fraud detection. It examines advances in model architectures, such as convolutional neural networks, long short-term memory networks, transformers, and ensemble methods—across domains including credit cards, insurance, and financial reporting. This review identifies key challenges in data imbalance, automation, explainability, and compliance with data privacy regulations. As a key contribution, this study introduces the deep learning–sector–governance framework, a novel synthesis that aligns DL methods with sector-specific constraints and regulatory requirements to guide the development of scalable and compliant fraud detection systems. By integrating methodological trends, regulatory shifts, and domain needs, this review offers practical guidance for researchers, industry practitioners, and policymakers working to strengthen fraud mitigation strategies in increasingly complex financial ecosystems.

1. Introduction

Financial fraud includes deceptive activities such as credit card misuse, insurance fraud, and money laundering which cause significant economic losses and undermine trust in financial institutions (Cao, 2022). According to global estimates, organizations lose approximately 5% of their annual revenue due to fraud, amounting to trillions of dollars each year. The proliferation of digital transactions, with over 2.7 billion in the United States alone in 2023 (Commission, 2024), has intensified the threat landscape, exposing new vulnerabilities and increasing the frequency and sophistication of fraud attempts.

Traditional fraud detection systems—relying on rule-based algorithms and manual audits are becoming increasingly inadequate (Zhu et al., 2024), often producing high false positives and delayed responses because of their inflexibility in adapting to evolving fraud tactics. In contrast, deep learning (DL) offers a transformative approach by leveraging high-dimensional data representations to uncover subtle, non-linear fraud signals. Models such as convolutional neural networks (CNNs), long short-term memory (LSTM) networks, and natural language processing (NLP) techniques are capable of real-time anomaly detection, enabling proactive and adaptive responses to fraud.

Furthermore, DL techniques are increasingly aligned with data protection laws such as the General Data Protection Regulation (GDPR) and California Consumer Privacy Act (CCPA), making them suitable for deployment in privacy-sensitive environments. These technologies not only enhance fraud detection capabilities but also align with the evolving legal landscape governing financial data use.

Despite their promise, DL applications for fraud detection face significant challenges. Key barriers include the need for high-quality labeled data, the complexity of interpreting model decisions, and ethical concerns regarding transparency and fairness (Bockel-Rickermann et al., 2023). Addressing these limitations requires a comprehensive understanding of the current trends, methodologies, and best practices in DL applications for fraud detection.

This paper contributes to the field by systematically reviewing 108 peer-reviewed studies published between 2019 and 2024, using the Kitchenham literature review framework. It identifies key technological advancements, evaluates the effectiveness of various DL models across

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the financial domains, and examines the influence of regulatory and privacy considerations. This study provides actionable insights for researchers, practitioners, and policymakers, aiming to bridge methodological innovation with real-world applicability in combating financial fraud.

To guide this analysis, the study explores the following research questions:

(1) What trends can be observed in the types of financial fraud addressed using DL, and how have data privacy, anonymization, and regulatory changes shaped these applications?
(2) How have feature engineering, data preprocessing, and automation, particularly in handling imbalanced data, improved the effectiveness and evaluation of DL models in financial fraud detection?
(3) What advancements have been made in DL models for financial fraud detection, and what trends are observed in their domain-specific applications?

These questions align with the objectives of this study and offer insights for researchers and practitioners to drive innovation in financial fraud detection.

Given this context, the 2019–2024 time frame was deliberately chosen to capture a pivotal period marked by several transformative developments in DL and financial fraud detection. This window followed the release of foundational models in 2018, significantly influencing the field. Notable examples include Zhang et al. (2018), who proposed a CNN-based model for online transaction fraud detection; Roy et al. (2018), who applied deep neural networks to credit card fraud; and Wang and Xu (2018), who combined DL with latent Dirichlet allocation (LDA)-based text analytics to detect automobile insurance fraud. These early contributions helped spread academic and industrial research, which intensified after 2019. Concurrently, the enforcement of major data privacy regulations—most notably the GDPR in 2018 and the CCPA in 2020—stimulated research into privacy-preserving machine learning (ML) methods, including federated learning, differential privacy, and blockchain-based compliance architectures. Additionally, the release and popularization of key domain-specific datasets, such as the European Credit Card dataset and synthetic insurance claims data (2019–2022), have enabled reproducible benchmarking and accelerated empirical research. Together, these technological, regulatory, and data-access inflection points make 2019–2024 a uniquely relevant and concentrated period for reviewing the role of DL in financial fraud detection.

This study makes three primary contributions to the field. First, it provides the most comprehensive review to date of DL-based financial fraud detection from 2019 to 2024, analyzing developments in model architecture, sector-specific adaptation, and regulatory considerations. Second, it introduces the deep learning–sector–governance (DLSG) framework, a novel conceptual model that integrates DL techniques, operational constraints unique to each financial sector, and legal governance requirements into a unified approach for developing scalable and regulation-compliant fraud detection systems. Third, this study translates its findings into practical guidance by synthesizing model development strategies, data governance practices, and deployment considerations, offering a holistic roadmap for both researchers and practitioners working to advance fraud mitigation in complex, data-driven financial ecosystems.

The paper is structured as follows. Section 2 reviews the existing literature on traditional and DL approaches for financial fraud detection, highlighting key challenges such as data imbalance, lack of real-world datasets, and the need for explainability and regulatory compliance. Section 3 outlines the research methodology, including study design, search and selection criteria, data extraction, and bias assessment techniques. Section 4 presents the results, addressing each of the research questions through data visualization and analysis. Section 5 introduces and develops the DLSG framework, which synthesizes the key findings across DL techniques, domain-specific fraud challenges, and data governance considerations. This framework provides a structured lens for understanding how technical innovation, the sector context, and regulatory compliance intersect in modern fraud detection systems. Section 6 discusses the limitations of the current review and proposes future research directions aligned with the DLSG framework, including actionable insights for practitioners of financial institutions, technology development, and regulatory governance. Section 7 concludes the study by summarizing the core insights and outlining the implications for both academic research and industry practice.

2. Literature review

Early studies on financial fraud detection primarily relied on traditional data mining techniques such as logistic regression (LR), neural networks, Bayesian belief networks, and decision trees (DTs). Ngai et al. (2011) and Batmaz et al. (2017) conducted seminal reviews of 49 studies published between 1997 and 2008, categorizing financial fraud detection research into sectors such as banking, insurance, and securities fraud. Their findings showed that classification and clustering methods dominated the early phase of research. Although effective in structured environments, these approaches often lack scalability and struggle to adapt to the growing sophistication and dynamism of fraudulent schemes, particularly in different financial sectors.

As the limitations of supervised methods have become increasingly apparent, particularly their dependence on labeled data, researchers have begun to explore anomaly detection and unsupervised learning techniques. A previous study (Ahmed et al., 2016) emphasized the growing importance of clustering-based anomaly detection in the financial domain, where access to real-world labeled datasets is often limited. These methods helped identify novel fraud patterns but faced challenges in validation and generalization, as many studies relied heavily on synthetic data. Akoglu et al. (2014) highlighted the emergence of graph-based anomaly detection, noting that graph structures capturing relationships between financial entities offer a richer representation of fraudulent behaviors. Their comprehensive framework distinguishes between static, dynamic, attributed, and plain graphs, showcasing the evolution of network-based approaches.

Recently, DL has gained prominence in financial fraud detection owing to its ability to model high-dimensional and complex data. According to Ozbayoglu et al. (2020), DL models such as CNNs, recurrent neural networks (RNNs), and transformers have outperformed classical ML models by capturing subtle, non-linear fraud patterns. Among the DL approaches, graph neural networks (GNNs) have emerged as particularly powerful. Motie and Raahemi (2024) conducted a systematic review of GNN applications in financial fraud detection and found that while most studies employed supervised or semi-supervised learning, significant opportunities remain for unsupervised GNN models capable of detecting previously unseen fraud behaviors.

However, despite these technical advances, several challenges persist. Financial fraud remains a rare event that leads to highly imbalanced datasets that impair model training and evaluation. Guo et al. (2017) extensively reviewed methods to address class imbalance, including oversampling techniques such as synthetic minority oversampling technique (SMOTE), ensemble learning, and cost-sensitive classification. Furthermore, the scarcity of real-world fraud data continues to limit external validity, and synthetic datasets often fail to capture the evolving complexity of fraud tactics. Another major challenge is the explainability of complex artificial intelligence (AI) models. Černevičienė and Kabašinskas (2024) emphasized the growing need for explainable AI (XAI) in finance, highlighting the importance of techniques such as SHapley Additive exPlanations (SHAP) values, feature importance analysis, and rule-based frameworks to meet regulatory demands and build stakeholder trust.

While recent studies have advanced model development and anomaly detection, a critical gap remains in the systematic integration of sector-specific requirements and governance considerations into fraud detection system design. Many studies focus on algorithmic innovation in isolation, often overlooked how sectoral factors, such as regulatory constraints, operational workflows, and fraud typologies unique to banking, insurance, or cryptocurrency, should inform model development and evaluation. Governance frameworks that address data privacy, compliance, and auditability have not yet been consistently embedded. Bridging these gaps is essential for translating technical advancements into scalable, practical, and trustworthy fraud-detection solutions.

Emerging trends suggest promising new directions but also reinforce the need for a more integrated approach. The incorporation of blockchain technology offers opportunities for enhancing transparency and security. Sharma et al. (2024) reviewed blockchain applications in finance and suggested that decentralized and immutable ledgers could significantly strengthen fraud prevention systems. Cognitive computing approaches, which combine ML with human expertise, have also been proposed to address the increasing complexity of fraud vectors. Ryman-Tubb et al. (2018) advocated greater collaboration between industry and academia to promote the operational deployment of AI in payment card fraud detection. Hilal et al. (2022) highlight a shift toward semi-supervised and unsupervised learning models, reflecting the need for techniques that function effectively with limited labeled data and adapt to emerging fraud patterns.

Finally, Bockel-Rickermann et al. (2023) synthesized insights from nearly 300 studies to organize fraud analytics research into distinct application domains and methodological categories. Their findings underline the pressing need for sector-specific customization and standardized benchmarks to ensure that models are not only technically accurate but also operationally relevant. Collectively, although the recent literature has made substantial strides in algorithmic innovation, it still leaves the challenge of systematically aligning technical methods with sector realities and governance frameworks.

This study addresses these critical gaps by introducing the DLSG framework. By synthesizing DL innovation, sector-specific operational needs, and regulatory compliance requirements, the DLSG framework offers a structured approach to designing fraud detection systems that are not only technically robust, but also contextually relevant and governance-aligned. In doing so, it aims to bridge the divide between methodological advancement and real-world applicability, ultimately strengthening the resilience and trustworthiness of financial fraud detection ecosystems.

3. Research method

3.1. Study design

This study adopts the Kitchenham review framework, which is known for its structured approach to evaluating advancements in dynamic fields. This facilitates a thorough analysis of the literature, uncovering gaps, trends, and challenges in applying DL to financial fraud detection. Key stages, including study selection, data extraction, and synthesis, were customized to address the field's interdisciplinary and algorithmic diversity. By ensuring transparency, replicability, and unbiased results, this framework provides a strong foundation for identifying opportunities and guiding future research.

3.2. Search criteria

A comprehensive search strategy was developed to identify relevant studies. Databases such as PubMed, SSRN, IEEE Xplore, ACM Digital Library, ScienceDirect, and Scopus were selected because of their strong coverage of interdisciplinary and technical research.

The search query was designed using Boolean operators ("AND", "OR") to combine keywords across three core domains: methodology, financial sectors, and fraud-related terminology. "AND" was used to ensure that retrieved studies addressed multiple relevant dimensions, narrowing the focus to financial fraud detection using ML and DL techniques. "OR" expanded the search within each domain to capture a wider variety of terms and phrasing used across different disciplines.

Specifically, the search included keywords related to methods (such as ML, AI, DL, and anomaly detection), financial domains (including banking, insurance, credit, tax, blockchain, and related sectors), and fraud-related terms (fraud, risk, and scam).

To refine the results and maintain relevance, the search excluded review and survey papers that focused exclusively on the original research. It also limited the subject areas to Computer Science, Business Management, Accounting, Economics, Finance, Decision Sciences, and Engineering to target studies positioned at the intersection of technology and finance.

The search process was carefully documented to ensure transparency and replicability. All search queries and results were managed using a reference management system to remove duplicates and to streamline the screening process.

3.3. Selection criteria

To ensure a rigorous and methodologically sound review, we applied detailed inclusion and exclusion criteria in alignment with Kitchenham's framework.

We limited the selection to peer-reviewed journal articles and conference papers published between 2019 and 2024, a period marked by a notable surge on DL research in financial fraud detection. Studies were included only if they (1) explicitly employed DL models such as CNNs, LSTMs, transformers, or GNNs; (2) addressed fraud detection in financial services (e.g., credit card, insurance, banking, cryptocurrency); and (3) provided empirical results supported by datasets, performance metrics, or reproducibility details. We excluded studies that were (1) not in English; (2) purely conceptual or theoretical without model implementation; (3) focused on non-financial domains (e.g., cybersecurity, identity theft); or (4) used only traditional ML methods without DL components.

This multi-stage filtering process starts with keyword-based retrieval, followed by abstract and full-text screening. It ensured that only methodologically relevant and domain-specific studies were retained. This approach enhances the transparency, objectivity, and reproducibility of our review process and helps ensure that the dataset reflects the current state-of-the-art DL-based financial fraud research.

3.4. Data extraction and analysis

To ensure consistency and comprehensiveness, a structured data extraction form was developed to standardize the collection and synthesis of data across studies. This enabled clear comparisons while minimizing subjectivity.

The data analysis utilized Python with libraries such as Pandas for data manipulation, Matplotlib for visualization, and Scikit-learn for ML and statistical tasks. We further visualized keyword co-occurrence network graphs, uncovering connections and trends among the key terms in the reviewed articles. This integrated approach ensured a robust and insightful analysis.

3.5. Risk of bias assessment

A structured approach was used to assess the potential sources of bias in the selected studies. Selection bias was minimized using predefined inclusion and exclusion criteria, as well as independent screening conducted by multiple authors. To address publication bias, this review included both peer-reviewed journal articles and preprints, ensuring a broader and more inclusive set of findings.

Performance bias was evaluated by examining variations in the data preprocessing steps, feature selection methods, and model evaluation procedures. Dataset bias was considered by assessing the limitations of publicly available datasets, particularly those issues related to class imbalance and representativeness. Reproducibility bias was assessed

based on how transparently each study reported algorithmic details, dataset availability, and evaluation methods.

Studies identified as having a high risk of bias were flagged during the synthesis to ensure that the final conclusions were grounded in well-documented and methodologically sound research. This assessment process is key to maintaining the objectivity, credibility, and overall reliability of the review.

The initial database search retrieved 2,858 records from multiple sources, including PubMed (47), SSRN (134), IEEE (98), Scopus (481), Taylor & Francis (404), and ScienceDirect (1,694). After removing duplicate entries, 2,514 unique records remained. Relevance screening was then applied to exclude studies unrelated to ML and financial fraud, thus reducing the dataset to 427 papers. Further refinement focused specifically on DL techniques, narrowing the selection to 108 studies.

To ensure the highest quality of the included studies, a rigorous quality assessment was conducted, prioritizing empirical studies with robust data analysis, reproducible methodologies, and peer-reviewed publications. Studies with limited experimental validation, insufficient dataset details, or a lack of methodological transparency were excluded. The final evaluation resulted in 108 high-quality studies that met the review standards and objectives. The selection process is illustrated in Fig. 1.

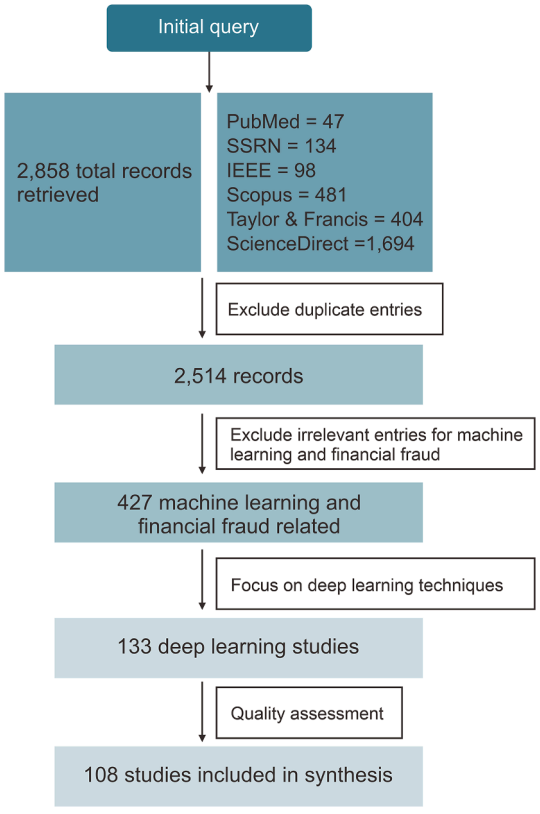


Fig. 1. Literature selection process for deep learning (DL) in financial fraud detection.

4. Results

4.1. Financial fraud types and privacy trends

Research question 1: What trends can be observed in the types of financial fraud addressed using DL, and how have data privacy, anonymization, and regulatory changes shaped these applications?

4.1.1. Yearly publication trends and sector focus

DL techniques have been applied across various financial fraud domains, with noticeable concentration in a few key sectors. As shown in Fig. 2, DL-based fraud research has grown rapidly since 2019, particularly since 2022. This trend coincides with the increasing scale and complexity of digital financial fraud, including online transaction fraud, cryptocurrency-related scams, and investment fraud, as reported by regulatory agencies such as the Federal Trade Commission (FTC) (2024).

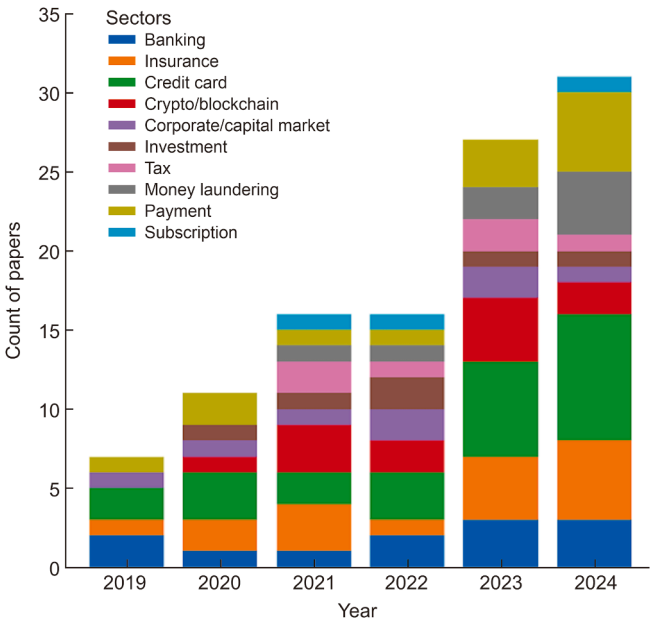


Fig. 2. Yearly trends in financial fraud research publications from 2019 to 2024.

Fig. 3 shows that credit card and banking fraud dominate this field. Credit card fraud is the most studied (Ryman-Tubb et al., 2018; Seera et al., 2024), reflecting the increasing awareness of fraud detection in credit card transactions. Public availability of credit card related datasets, such as the European Credit Card Transactions dataset Kaggle has likely contributed to this trend, as it provides researchers with clean and standardized data to develop and evaluate state-of-the-art DL models.

Furthermore, the banking and insurance sectors show high focus in relevant research, emphasizing the growing need to tackle fraud in digital payments (Faridpour and Moradi, 2020; Jurgovsky et al., 2018; Kuzey et al., 2014; Tekkali and Natarajan, 2023), automobile insurance claims (Jaiswal et al., 2024; Ming et al., 2024; Settipalli and Gangadharan, 2023), health insurance claims (Kose et al., 2015; Sowah et al., 2019).

Emerging areas such as crypto/blockchain and payment systems show a notable number of studies, indicating an increasing focus on fraud prevention in digital currencies (Alarab et al., 2020). However, sectors such as taxes, mortgages, loans, and money laundering have relatively few publications, which could be due to limited access to domain-specific datasets or the complexity of detecting fraud patterns in these areas (Blaszczynski et al., 2021; Kim and Kang, 2016; Liu et al., 2022; Matos et al., 2020; Tonnelier et al., 2018; Vanhoeyveld et al., 2020).

According to Consumer Sentinel Network Reports published by the Federal Trade Commission, bank transfers and payments accounted for the highest aggregate losses reported in 2023 ($1.86 billion), followed closely by Cryptocurrency ($1.41 billion), while credit cards were most frequently identified as the payment method in fraud reports. Additionally, the number of insurance-related identity theft reports increased by 26% compared with the previous year (Commission, 2024). The sheer scale and rapid growth of financial fraud across these sectors have significantly driven the observed surge in research interest and innovation in this field.

These substantial financial losses, particularly in sectors such as banking, cryptocurrency, credit cards, and insurance—not only

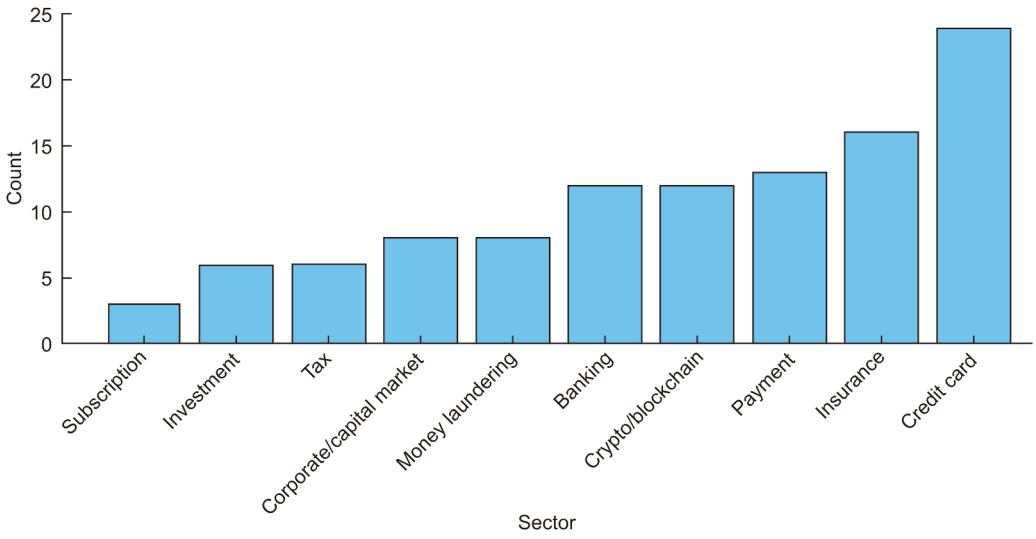


Fig. 3. Financial sector trends in financial fraud detection research papers.

highlight the urgency of effective fraud detection but also underscore the increasing scrutiny of how sensitive data is handled. As fraudulent activities grow in scale and complexity, regulatory bodies have intensified their focus on data privacy and compliance, placing additional demands on fraud detection systems. In response, researchers have begun to integrate privacy-preserving techniques and comply with frameworks such as the GDPR and CCPA, fundamentally shaping the development of DL models for fraud detection.

4.1.2. Influence of data privacy, anonymization, and regulation

Of the 108 papers reviewed, 23 had publicly available datasets and 85 used non-public datasets, and 67 addressed privacy issues.

Principal component analysis (PCA) enhances data privacy in fraud detection by reducing high-dimensional data into lower-dimensional representations and preserving critical information while minimizing information loss. This transformation anonymizes transactional data, removes identifiable personal information and enables secure sharing across institutions without exposing sensitive details. Additionally, PCA aids compliance with data protection regulations by mitigating re-identification risks and retaining only essential features needed for fraud detection, thus reducing unnecessary exposure to sensitive transaction details. However, by altering the original features, PCA can reduce model interpretability, potentially limiting insight into the relationships between variables (Vouk et al., 2023). However, its ability to balance privacy and performance makes it a valuable tool for secure data processing.

Blockchain enhances data privacy in financial fraud detection owing to its decentralized and immutable nature, eliminating the need for centralized intermediaries (Devaguptam et al., 2024; Mohammed et al., 2024). Sensitive data, such as personal identities and financial transactions, is securely distributed across an encrypted ledger, thereby preventing unauthorized access. Pseudonymity is ensured by cryptographic addresses, which further safeguard personal information.

Blockchain also enables real-time privacy compliance through smart contracts and automates data-sharing policies while maintaining confidentiality (Krishnan et al., 2023). For instance, in cryptocurrency fraud detection, selective data disclosure minimizes information exposure during verification, aligning with regulations such as the GDPR (Ahmad and Aujla, 2023). These features make blockchain vital for secure and compliant fraud detection systems.

Both the GDPR and CCPA significantly impact the application of ML in financial fraud detection by enforcing strict data privacy, security, and transparency requirements. The GDPR mandates organizations to obtain consent, anonymize sensitive data, and adhere to ethical standards, emphasizing transparency, accountability, and data minimization (Mwangi, 2024). Similarly, the CCPA grants individuals the right to access, delete, or opt out of data usage, requiring ML models to anonymize or pseudonymize data to protect identities.

A critical GDPR mandate is the "right to explanation", which ensures that AI decisions, such as fraud detection outcomes, are interpretable (Brughmans et al., 2024; Cao et al., 2018; Lin and Gao, 2022; Kurshan et al., 2020; Vouk et al., 2023). This presents a challenge for complex models like deep neural networks, prompting the development of explainability techniques to balance compliance and performance (Chen and Han, 2023; Hyun and Shin, 2022; Kloska et al., 2023). Similarly, CCPA emphasizes consumer control and data minimization, posing challenges for ML systems reliant on extensive datasets. Techniques such as data anonymization, federated learning, and secure multiparty computation have become essential to comply with both regulations. Non-compliance risks penalties under both frameworks, driving organizations to invest in secure, transparent, and privacy-preserving ML systems, which ultimately enhance trust, scalability, and effectiveness in fraud detection (Mavundla et al., 2024; Shen and Kurshan, 2021).

The Council of Health Insurance (CHI) (2024) regulates private health insurance in Saudi Arabia, ensuring fraud prevention, data integrity, and healthcare accessibility. Under Saudi Vision 2030, the CHI mandates detailed records of fraud, fostering transparency and accountability among insurers and providers.

CHI's initiatives, such as the National Platform for Health and Insurance Exchange Services (NPHIES) (2024), enhance data security and interoperability. By leveraging advanced technologies, including ML, the platform improves fraud detection and processing efficiency, aligning with goals to reduce fraud costs and comply with data protection and ethical standards in health insurance.

4.2. Preprocessing, imbalance handling, and evaluation

Research question 2: How has feature engineering, data preprocessing, and automation—particularly in handling imbalanced data, improved the effectiveness and evaluation of DL models in financial fraud detection?

Financial fraud detection faces significant challenges because of imbalanced datasets in which fraudulent transactions represent a small fraction of the total (Guo et al., 2017; Kaur et al., 2019; Tian et al., 2022;

Vandervorst et al., 2022; Wang et al., 2023a). Of the 108 reviewed papers, 92 reported dataset imbalance issues, leading traditional ML models to favor the majority class and underperform in fraudulent cases. Addressing this issue requires advanced preprocessing and automation techniques.

To address these limitations, researchers have developed a range of preprocessing, automation, and evaluation strategies, as discussed in the following subsections.

4.2.1. Preprocessing techniques

Oversampling methods, such as the SMOTE, generate synthetic samples by interpolating minority class instances, thereby effectively enhancing class diversity and model robustness (Chawla et al., 2002; Ming et al., 2024; Nasir et al., 2021; Wang et al., 2024). SMOTE has also proven to be effective for handling imbalanced datasets with missing values.

Stratified sampling is a probabilistic technique that divides data into strata based on timestamps to address non-stationary changes in transaction fraud characteristics. This ensures a higher representation of recent fraudulent transactions while maintaining randomness (Charizanos et al., 2024).

Advanced approaches such as generative adversarial networks (GANs) and variational autoencoders (VAEs) produce realistic synthetic data while maintaining the original distribution, further supporting model training in extreme class imbalance scenarios. Data imputation techniques, such as k-nearest neighbors (kNN), address missing values to improve overall data quality.

Feature transformations, including scaling and normalization, play a critical role in reducing biases and enhancing fraud pattern detections. Techniques such as adaptive synthetic sampling (ADASYN) focus on harder-to-classify samples whereas cluster-based oversampling generates synthetic data tailored to domain-specific contexts, ensuring alignment with real-world complexities.

4.2.2. Automation techniques

Automation significantly accelerates fraud detection by reducing manual intervention and enabling substantial cost savings (Aslam et al., 2022). Subsampling techniques effectively select representative datasets and minimize computational costs while preserving feature correlations (Siddamsetti and Srivenkatesh, 2024).

Blockchain enables decentralized, secure data sharing and automated fraud detection via smart contracts, ensuring rapid responses and model adaptability (Alshawi, 2023; Kaafarani et al., 2024). Techniques such as stochastic gradient descent (SGD) optimize models incrementally, whereas automated parameter tuning (Hofmann et al., 2022; Wang et al., 2023b; Zhao, 2024) and resampling hybrid techniques (Abdul Salam et al., 2024) enhance the detection accuracy.

Knowledge distillation transfers insights from complex models to lightweight models, enabling efficient real-time detection. Fraud detection pipelines streamline processes by combining preprocessing and model training into cohesive systems. Stacking ensembles improve robustness by combining classifiers (Louzada and Ara, 2012; Zhang et al., 2022, 2024), with methods such as the random forest quantile classifier optimizing sensitivity and specificity for imbalanced data (Carracedo et al., 2024). These advancements ensure scalable, adaptable, and accurate fraud detection systems capable of addressing the complexities of financial fraud.

Although automation enhances detection scalability and reduces latency, the effectiveness of these models ultimately depends on how their performance is evaluated, particularly in the context of class imbalance and operational risk.

4.2.3. Evaluation metrics

Traditional metrics face limitations in handling imbalanced datasets where fraud cases are rare (Fernandez et al., 2022). Metrics such as accuracy are often misleading, favoring the majority class and overlooking fraud. Precision minimizes false positives but may reduce recall, which identifies actual fraud cases. High recall, while critical, can increase the false positive rate (FPR) and operational costs. The F1-score balances precision and recall but does not address economic impacts.

Metrics such as the false negative rate (FNR) and FPR highlight specific challenges, such as losses from undetected fraud or inefficiencies from false alarms (Zhao et al., 2016). Unlike area under the receiver operating characteristic curve (AUC-ROC), which may be inflated owing to class imbalance, area under the precision-recall curve (AUC-PR) provides a clearer picture of performance in the minority class by focusing only on precision and recall, making it more appropriate for fraud-detection tasks. The metrics and their formulas are shown in Table 1.

Economic metrics, such as the cost of FPs and cost of FNs, assess operational and financial impacts, helping organizations optimize detection strategies. For instance, precision is prioritized in cryptocurrency fraud to minimize compliance costs, while recall is critical in tax fraud to prevent revenue losses.

In conclusion, effective fraud detection increasingly relies on tailored metrics that address imbalanced datasets, operational costs, and sector-specific needs, complementing foundational metrics such as accuracy and F1-score with domain-specific benchmarks.

4.3. Model advancements in financial fraud detection

Research question 3: What advancements have been made in DL models for financial fraud detection?

For advancements in DL models, we identified the following DL models, ML models, and hybrid models, which are widely used in fraud detection. Traditional ML models such as logistic regression, random forest, and gradient boosting are often used as benchmarks, especially for structured data and less computational intensity; however, they are often less capable to capture non-linear relationships (Ngai et al., 2011; Ozbayoglu et al., 2020).

DL models such as CNNs and RNNs can extract high-dimensional data patterns (Wang et al., 2019). Sequential models including RNNs, LSTM and gated recurrent units (GRU) are adept at modeling sequential transaction histories (Jurgovsky et al., 2018). Transformers outperform traditional sequence models by capturing long-range dependencies through self-attention, making them highly effective in large-scale behavioral modeling.

To identify fraud rings, GNN models relationships between users, devices, and transactions (Cheng et al., 2023; Hu et al., 2022; Ma et al., 2023; Motie and Raahemi, 2024; Zhou et al., 2017). Meanwhile, generative models such as GANs and VAEs help address class imbalance by creating synthetic data (Aftabi et al., 2023). Hybrid models, such as transformer-local outlier factor (LOF)-random forest (RF), autoencoder-LSTM, or CatBoost-deep neural network (DNN)—combine multiple architectures and assign weights to the model output. These hybrid approaches are effective in addressing imbalanced datasets, noisy inputs, and evolving fraud tactics, offering a balanced trade-off between accuracy, interpretability, and computational efficiency.

4.3.1. DL models

CNNs: By analyzing high-dimensional features such as time-series embeddings and transaction heatmaps, CNNs identify anomalies linked to unusual spending patterns or merchant-specific risks. They do not require heavy data preprocessing during training because it inherently captures key features and performs feature dimension reduction. It was found that deep features extracted from CNNs enhance fraud detection when combining CNN with traditional models (Ming et al., 2024).

RNNs: RNNs process sequential data by retaining the context from previous inputs, making them useful for analyzing patterns in transaction histories. However, their capacity is limited by gradient vanishing and exploding issues. The GRU and LSTM are specialized RNN and both

Table 1
Evaluation metrics for fraud detection.

Metrics name	Formula	Description
Accuracy	$\frac{TP+TN}{TP+TN+FP+FN}$	Measures the proportion of correctly classified instances (both fraud and non-fraud) out of the total instances. High accuracy is not always reliable in fraud detection due to class imbalance.
Precision	$\frac{TP}{TP+FP}$	Measures the proportion of correctly predicted fraud cases out of all predicted fraud cases. High precision indicates fewer FPs (legitimate transactions flagged as fraud).
Recall (Sensitivity)	$\frac{TP}{TP+FN}$	Measures the proportion of actual fraud cases correctly identified by the model. High recall indicates fewer FNs (fraudulent transactions missed by the model).
F1-score	$2\times\frac{Precision\times Recall}{Precision+Recall}$	The harmonic mean of precision and recall. It balances precision and recall, making it useful when both FPs and FNs are costly.
AUC-ROC	Area under the receiver operating characteristic curve (TPR vs. FPR)	TPR is plotted against FPR at various thresholds. AUC-ROC measures the model's ability to distinguish between fraud and non-fraud cases. Higher AUC-ROC indicates better performance.
AUC-PR	Area under the precision-recall curve (precision vs. recall)	Precision is plotted against recall at various thresholds. AUC-PR is especially useful for imbalanced datasets (common in fraud detection) as it focuses on the performance of the positive class (fraud). Higher AUC-PR indicates better performance.
FPR	$\frac{FP}{FP+TN}$	Measures the proportion of legitimate transactions incorrectly flagged as fraud. Lower FPR is desirable to reduce customer inconvenience.
FNR	$\frac{FN}{FN+TP}$	Measures the proportion of fraudulent transactions missed by the model. Lower FNR is critical in fraud detection to minimize financial losses.
Cost of FPs	$C_{FP}\times FP$	Represents the financial or operational cost associated with incorrectly flagging legitimate transactions as fraud (e.g., customer dissatisfaction, manual review costs).
Cost of FNs	$C_{FN}\times FN$	Represents the financial loss or risk associated with failing to detect fraudulent transactions (e.g., chargebacks, lost revenue). This is typically higher than the cost of FPs in fraud detection.

Note: TP: true positive; TN: true negative; FP: false positive; FN: false negative; AUC-ROC: area under the receiver operating characteristic curve; AUC-PR: area under the precision-recall curve; TPR: true positive rate; FPR: false positive rate.

GRU and LSTM mitigate the issues of gradient vanishing and exploding in RNNs (Hosseinnia Shavaki and Ebrahimi Ghahnavieh, 2023; Manirih et al., 2024; Zhu et al., 2023).

Multilayer perceptrons (MLPs): MLPs, composed of interconnected layers, model relationships in structured data. MLP is a competitive choice for fraud detection, particularly in amount-based profiling and scenarios requiring the handling of non-linear relationships in data.

Transformers: Unlike CNNs and RNNs, which process all points in the input sequence step by step, transformers process all points simultaneously. The self-attention mechanism and feed-forward networks of transformers enable it to model complex relationships and extract meaningful features from sequential data, which is useful for recognizing patterns in transactional data, user behavior, or network interactions in fraud detection.

NLP: NLP enhances fraud detection by analyzing unstructured textual data such as financial reports (Zhang et al., 2022), tax compliance and financial regulation, and claims narratives (Gonzalez-Martel et al., 2021). Key techniques include sentiment analysis, readability metrics, and feature extraction (e.g., bag-of-words, term frequency-inverse document frequency, and word embeddings). These features are integrated with traditional fraud indicators in ML models to improve accuracy and recall (Craja et al., 2020; Humpherys et al., 2011; Wang and Xu, 2018; Zhang et al., 2022).

VAEs and GANs: GANs and VAEs are two popular types of generative models used in DL (Jiang et al., 2023). GANs excel in generating high-quality, realistic samples but are harder to train owing to adversarial dynamics and lack of an interpretable latent space. VAEs are better for representation learning and probabilistic modeling, with a well-structured latent space, but they generate samples of lower quality compared to GANs. Combining the strengths of VAEs and GANs can address the limitations of handling imbalanced data for fraud detection.

GNNs: GNNs are highly effective in fraud detection because they can model complex interactions and relationships between entities (e.g., transactions and reviews) as graphs (Akoglu et al., 2014; Deng et al., 2022; Innan et al., 2024; Wang et al., 2023a). Fraud detection often involves analyzing these relationships to identify patterns and anomalies indicative of fraudulent activity (Khodabandehlou and Golpayegani, 2022; Nguyen et al., 2023; Shehnepoor et al., 2024). Graph convolutional networks (GCNs) are a specific type of GNN that apply the concept of convolution to graphs. Studies have shown that GCNs outperform traditional models such as logistic regression, random forest, and gradient boosting machines in detecting fraudulent transactions (Colladon and Remondi, 2017; Usman et al., 2023; Xu et al., 2023).

4.3.2. ML models

Machine learning models are commonly used as either standalone benchmarks or integral components of hybrid systems in financial fraud detection. Traditional ML algorithms serve as essential benchmarks for financial fraud detection (Alexandre and Balsa, 2023; Hilal et al., 2022). LR offers simplicity and interpretability, and is ideal for binary classification (Alexandre and Balsa, 2023; Hilal et al., 2022; Khan et al., 2022). LinearSVC efficiently handles linear data, whereas kNN detects anomalies based on proximity, although it lacks scalability for large datasets.

Ensemble methods such as RFs and gradient boosting (XGBoost, LightGBM) improve accuracy by combining multiple models. RFs resist overfitting, while gradient boosting effectively handles imbalanced datasets effectively. Adaptive boosting (AdaBoost) iteratively strengthens weak classifiers to address iteratively for fraud-specific challenges.

Support vector machines (SVMs) excel in high-dimensional spaces, identifying outliers effectively (Farvaresh and Sepehri, 2011). Decision trees, though prone to overfitting, remain interpretable and foundational for ensemble models (Dutta et al., 2017).

Although less adaptive than DL, these algorithms provide valuable benchmarks, offering insights into structured data performance and guiding advancements in fraud detection systems.

4.3.3. Hybrid models

Hybrid models integrate the complementary strengths of different algorithms to address complex tasks , such as financial fraud detection (Cao et al., 2023; Taha and Hadi, 2019; Zhai et al., 2017; Zhao et al., 2022; Zhu et al., 2025).

Adaptive sampling and aggregation-based graph neural network (ASA-GNN) enhance traditional GNN frameworks by integrating adaptive sampling and entropy-based aggregation, which addresses the major limitations of standard GNNs in handling fraud detection. By focusing on relevant neighbors and combating oversmoothing, the ASA-GNN provides a robust and scalable solution for identifying complex fraud patterns in graph-structured data (Tian et al., 2024).

Reinforcement learning with deep Q-network (RDQN) integrates DNN and reinforcement learning (Q-learning). By leveraging the rough

set theory for feature reduction and employing reinforcement learning, the RDQN model achieves faster processing and higher accuracy, making it a scalable and effective solution for real-world fraud detection problems. RDQN outperforms traditional models such as SVM, ANN, and DT, as well as hybrid models such as IFDTC4.5, SAE-GAN, and CNN-SVM-kNN (Tekkali and Natarajan, 2023).

The transformer-LOF-RF model uniquely combines the transformer's advanced feature extraction, LOF local anomaly detection, and RF ensemble learning to effectively detect complex and rare fraudulent patterns. It surpasses state-of-the-art models such as XGBoost, LightGBM, and LSTM by addressing data imbalance, reducing FPs and FNs, and adapting to emerging fraud techniques (Miao, 2024).

CatBoost-deep neural networks combine CatBoost and DNN to leverage their respective strengths. CatBoost excels at handling categorical features, imbalanced datasets, and complex relationships in structured data (Lokanan and Maddhesia, 2025). DNN focuses on learning patterns in raw features and adapting them to sparse data conditions. The hybrid model significantly outperforms other methods such as RFs and ensemble methods such as LSTM-based AdaBoost.

The autoencoder-LSTM is a combination of two deep learning models: autoencoder and LSTM. The autoencoder performs dimensionality reduction while retaining the key features and removing noise. The LSTM network models temporal dependencies and classifies transactions as either fraudulent or legitimate. The combined model demonstrated superior performance over traditional ML methods and standalone LSTM models (Zioviris et al., 2024).

4.3.4. Trends in model usage

The overall pattern of the frequency of application of the deep learning models is shown in Fig. 4. LSTM, MLP, CNN and RNN are most widely used. NLP methods such as bidirectional encoder representations from transformer (BERT) model are moderately applied to textual datasets. In addition, more specialized methods have been applied to detect fraud, including GNNs, GANs, and VAEs. As fraud detection datasets are highly imbalanced, whereas GANs are moderately applied, their potential in modeling relationships and generating synthetic datasets for fraud detection could drive more research in these areas.

The temporal evolution of research in DL for financial fraud detection reveals notable trends in both volume and methodological preference. As shown in Fig. 2, scholarly output increased substantially from 2019 onward, with a significant acceleration after 2022. This growth aligns with rising public concern over digital fraud, increased data availability, and intensified regulatory scrutiny (e.g., GDPR, CCPA). Furthermore, Fig. 5 demonstrates that models like LSTM and transformers have gained increasing popularity over time, likely due to their superior performance in handling sequential and unstructured data, which became more prevalent in financial applications after 2021. These shifts suggest that the field has moved from relatively simple architectures (e.g., CNNs and MLPs) toward more context-aware and adaptive models, reflecting the dynamic nature of fraud detection challenges and the growing need for real-time, interpretable systems.

As shown in Fig. 6, the most commonly used baseline ML algorithms in financial fraud detection studies are RF, ensemble learning methods, and LR. Their frequent application reflects both their widespread availability in standard ML libraries and their ease of implementation. These models also offer relatively strong baseline performance, which makes them practical choices for initial experimentation and benchmarking. However, traditional ML algorithms often struggle with imbalanced datasets which are prevalent in financial fraud scenarios, thus limiting their effectiveness in identifying rare but critical fraudulent events. This challenge is further explored in a later section of the paper.

Similarly, as shown in Fig. 7, there has been a notable increase in the use of RF, ensemble learning methods, and LR from 2019 to 2024, with a particularly sharp increase observed in 2023 and 2024. This trend suggests a growing reliance on these models as baseline approaches in financial fraud detection, likely because of their interpretability, accessibility, and proven performance in preliminary experiments.

Fig. 8 shows the distribution and frequency of various DL algorithms applied to financial fraud detection across different financial sectors. Both the credit card and banking sectors have significant applications of a wide range of deep learning techniques. In particular, the MLP, LSTM, and CNN are more commonly used. LSTM and GNN show significant application, likely owing to the sequential and graph-structured nature of blockchain data, which requires methods that can model relationships between transactions. Sectors such as tax and money laundering show minimal application of DL techniques.

To explore the interconnection between various methods and domains, this study conducted semantic network analysis, as shown in Fig. 9.

As shown in Fig. 9, Cluster 1 (colored in red) has 10 keywords related to ML models, including RF, LR, SVM, ensemble, DT. RF (with 32 links and 234 link strengths) and LR (with 31 links and 191 link strength) are the two most relevant keywords. These two traditional ML models are most frequently connected with keywords representing DL, indicating that they are often used as baselines or hybrid models.

Cluster 2 (colored in green) has 10 keywords related to DL models, including transformer, BERT, RNN. MLP (27 links and 103 link strengths) was the most relevant keyword. It is well connected to CNN, LSTM, SVM and RF.

Cluster 3 (colored in blue) contains seven keywords related to lightweight models, including Naive Bayes, kNN and DTs. This smaller cluster represents the lightweight models used for simple datasets.

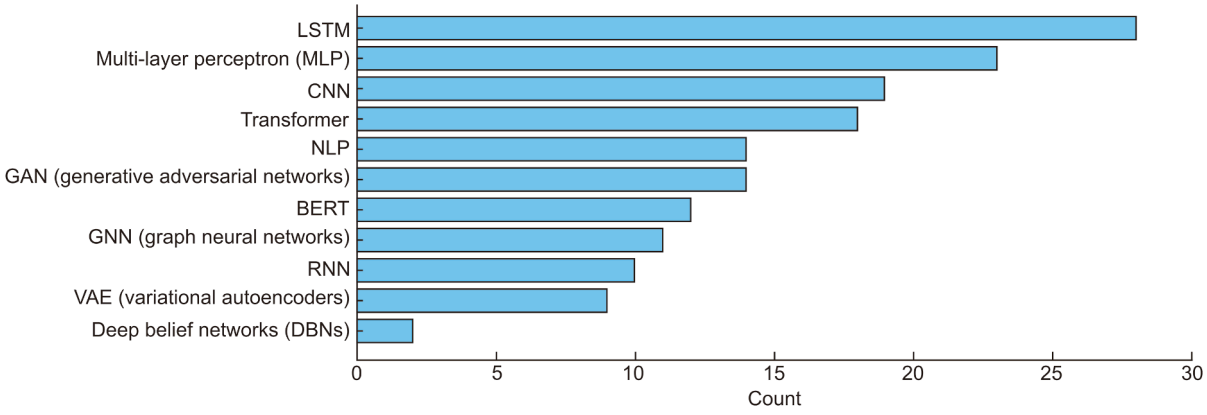


Fig. 4. Distribution of deep learning (DL) techniques applied to financial fraud detection. LSTM: long short-term memory; CNN: convolutional neural network; NLP: natural language processing; BERT: bidirectional encoder representations from transformer; RNN: recurrent neural network.

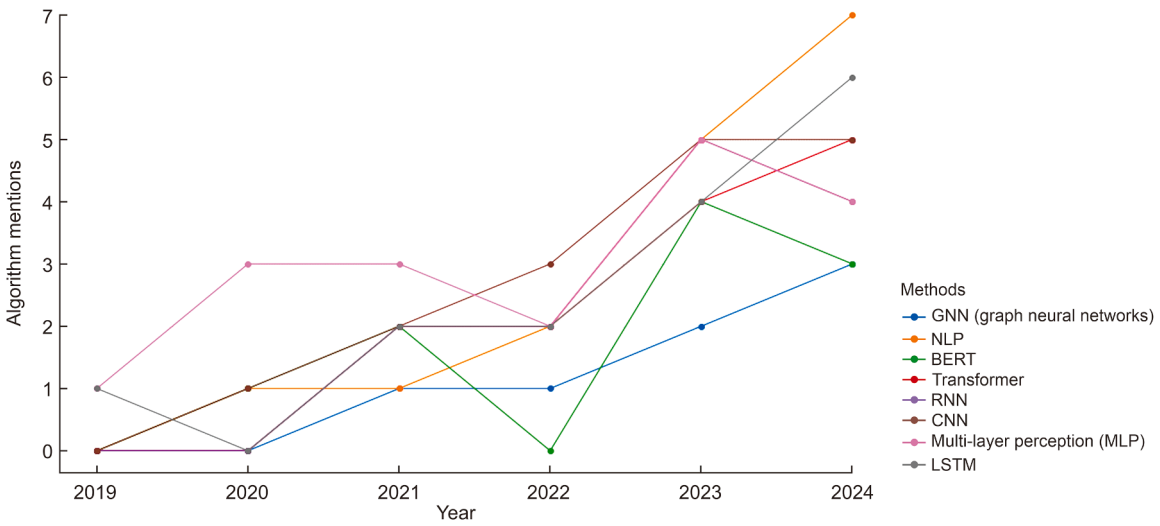


Fig. 5. Yearly trends of deep learning (DL) algorithm application in fraud detection from 2019 to 2024. NLP: natural language processing; BERT: bidirectional encoder representations from transformer; RNN: recurrent neural network; CNN: convolutional neural network; LSTM: long short-term memory.

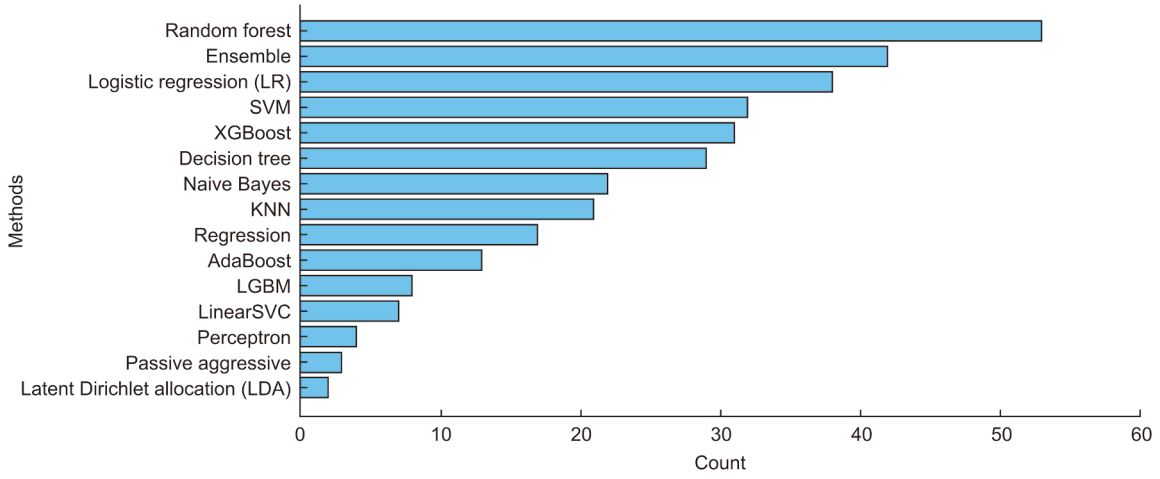


Fig. 6. Distribution of machine learning (ML) techniques applied to financial fraud detection. SVM: support vector machine; KNN: k-nearest neighbors; LGBM: light gradient boosting machine.

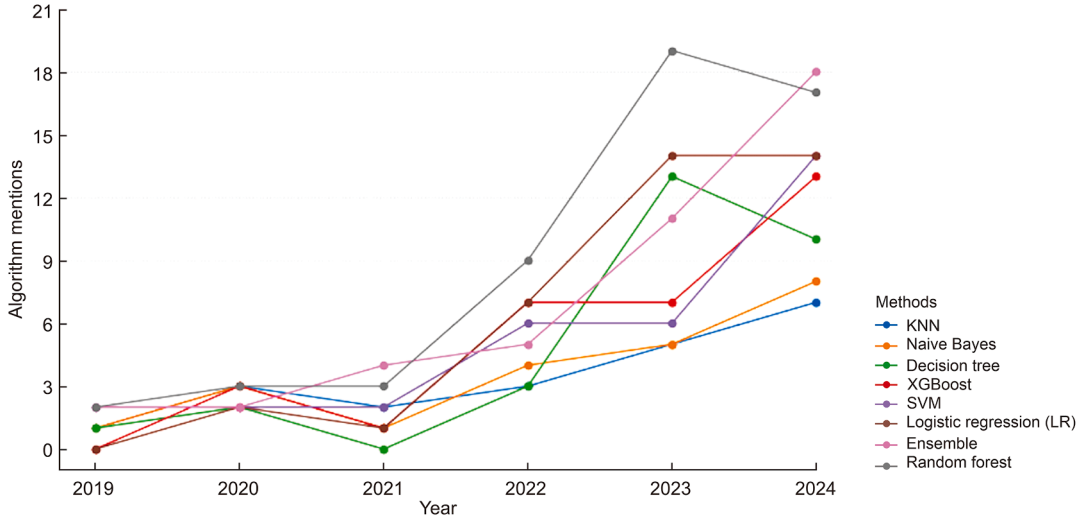


Fig. 7. Yearly trends of machine learning (ML) algorithm application in fraud detection from 2019 to 2024. KNN: k-nearest neighbors; SVM: support vector machine.

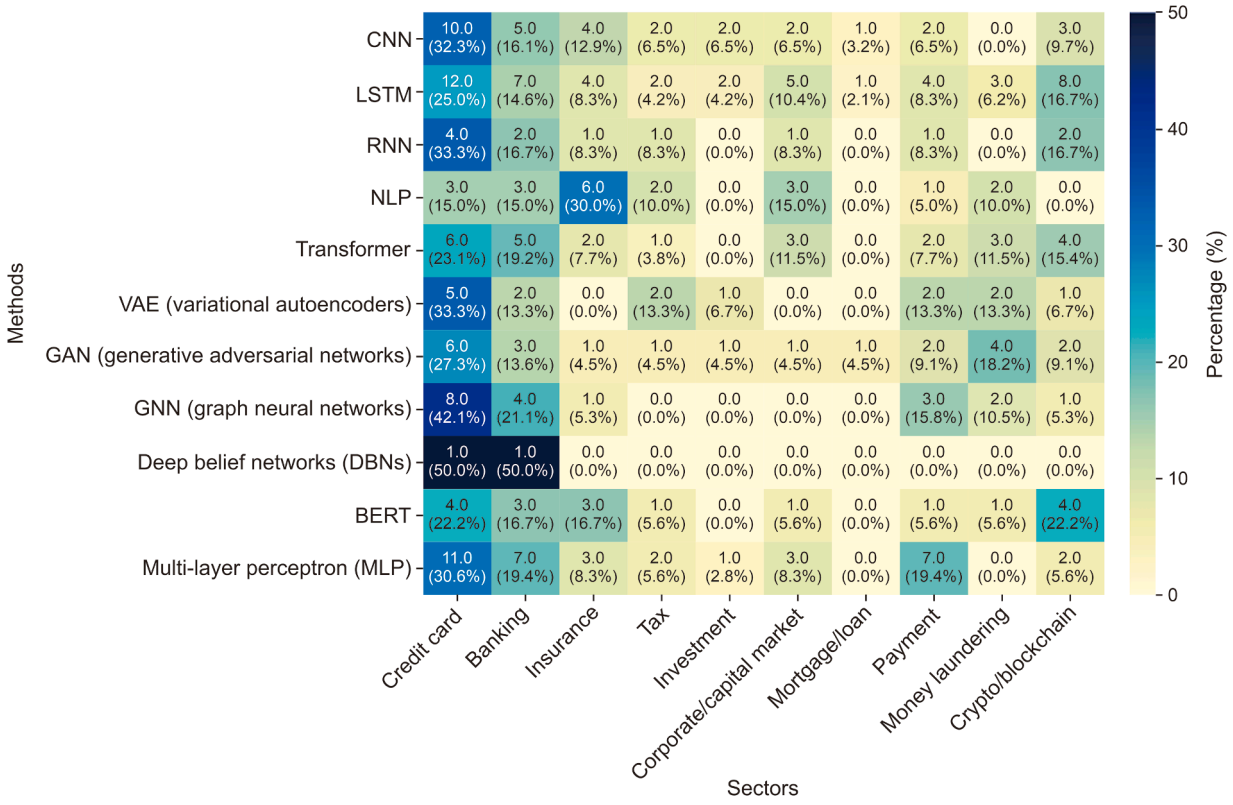


Fig. 8. Application of deep learning (DL) methods in financial fraud detection across sectors. CNN: convolutional neural network; LSTM: long short-term memory; RNN: recurrent neural network; NLP: natural language processing; BERT: bidirectional encoder representations from transformer.

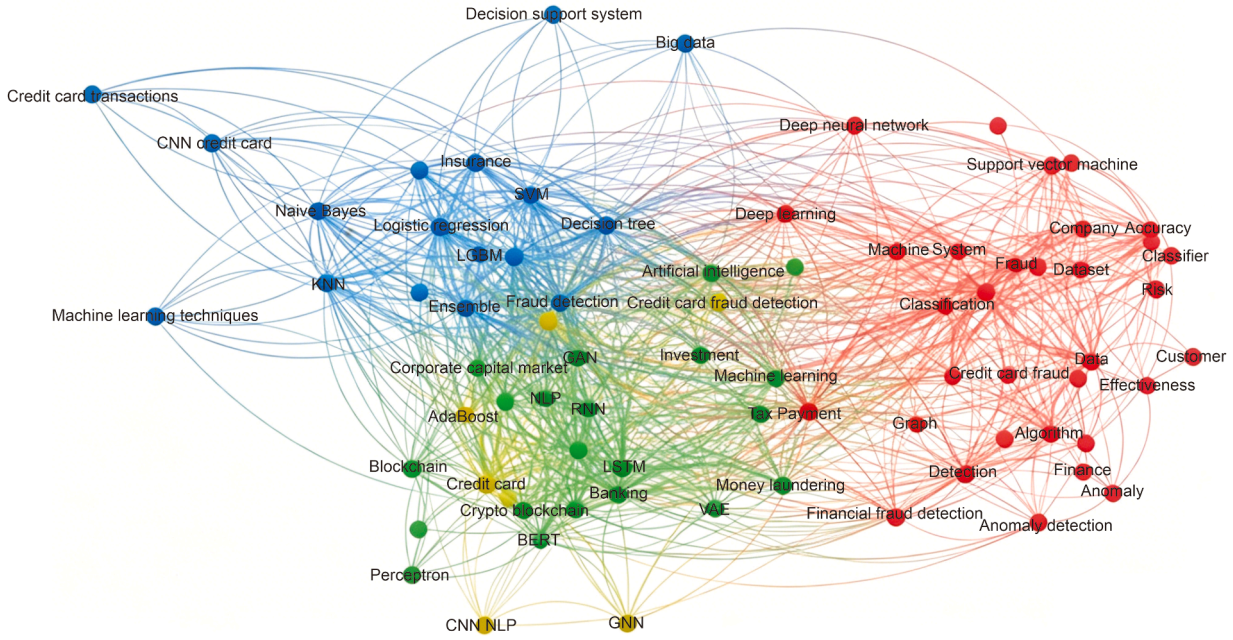


Fig. 9. Keyword co-occurrence network graph. CNN: convolutional neural network; LSTM: long short-term memory; RNN: recurrent neural network; NLP: natural language processing; BERT: bidirectional encoder representations from transformer; KNN: k-nearest neighbors; SVM: support vector machine; LGBM: light gradient boosting machine; VAE: variational autoencoder; GAN: generative adversarial network.

Cluster 4 (colored in yellow) has six keywords related to both the financial sectors and DL models. The credit card keyword is the most relevant node, with 30 links and a total link strength of 160. It also includes DL algorithms such as CNN and GAN, indicating the popular trend of CNN and GNN in fraud detection, especially in credit cards and banking.

Cluster 5 (colored in purple) has four keywords related to more specialized DL models such as GAN and VAE. This shows an emerging trend of applying GAN to generate synthetic data to address imbalanced datasets.

In summary, classical ML methods (e.g., RF and SVM) remain fundamental for structured data fraud detection. Techniques such as LSTM, Transformers, and BERT have been increasingly adopted for

analyzing sequential and text-based fraud data. GNNs and GANs have been gaining traction in specialized fraud domains like blockchain and synthetic data generation (Ahmed et al., 2016). Different algorithms are tailored to the needs of specific financial sectors (e.g., CNN for credit card fraud, GNN for blockchain).

5. The DLSG framework: integrating DL, sector context, and governance

To guide future research and development in financial fraud-detection, this study proposes the DLSG framework, a theoretical construct that synthesizes insights from DL innovation, sector-specific operational contexts, and data governance requirements.

Unlike prior frameworks, which often prioritize technical performance or regulatory compliance in isolation, the DLSG framework uniquely and systematically integrates DL innovation, sector-specific operational constraints, and data governance principles as co-equal pillars. This triadic alignment forms a theoretically grounded and operationally robust architecture that ensures that AI-driven fraud-detection systems are simultaneously scalable, interpretable, and regulation-ready. The DLSG not only synthesizes but strategically calibrates each component based on the dynamic needs of the domain such as ultra-low latency in credit card processing or federated data integrity in insurance, thereby filling a critical gap in existing frameworks. Moreover, each dimension is operationalized through clearly defined implementation layers: algorithm-level modules (e.g., LSTM + CNN hybrids), infrastructure support (e.g., edge computing or federated architectures), and governance enablers (e.g., explainability dashboards or GDPR audit trails), enabling a granular and actionable system design path.

As fraudulent activity grows in scale and complexity, particularly across digital financial systems, a robust and flexible framework is essential for aligning technical advancement with legal and organizational constraints.

As shown in Fig. 10, at the foundation of the DLSG framework lies DL, a transformative force in fraud-detection. Advanced models such as CNNs, LSTMs, GNNs, and transformer-based architectures have enabled the detection of subtle, high-dimensional fraud patterns that traditional rule-based systems often miss. These models are further enhanced through data-centric techniques such as SMOTE, ADASYN, and feature scaling, which address class imbalance and improve model generalization. Automation strategies—ranging from model ensembling and knowledge distillation to subsampling pipelines—reduce latency and cost while maintaining scalability. For instance, in credit card fraud-detection, CNNs and LSTMs support real-time decision-making by processing streaming transaction data (Carneiro et al., 2017; Charizanos et al., 2024), whereas hybrid models increasingly bridge performance and interpretability gaps across structured and unstructured data environments.

Beyond algorithmic sophistication, effective fraud-detection requires alignment with the sector-specific conditions in which these systems operate. Each financial domain such as credit cards, insurance, blockchain, and banking, exhibits distinct threats, workflows, and regulatory challenges. Credit card fraud-detection, for instance, demands exceptionally high precision and near-instantaneous response to prevent consumer disruption, and benefits from extensive access to publicly available datasets. In contrast, the insurance sector involves distributed data across patients, providers, and payers, requiring federated learning frameworks to enable collaborative fraud-detection while maintaining compliance with the Health Insurance Portability and Accountability Act (HIPAA) and GDPR regulations (Devaguptam et al., 2024). Blockchain-based financial services present unique challenges related to pseudonymity and transaction immutability; here, permissioned blockchains and smart contracts enable secure data sharing and automated

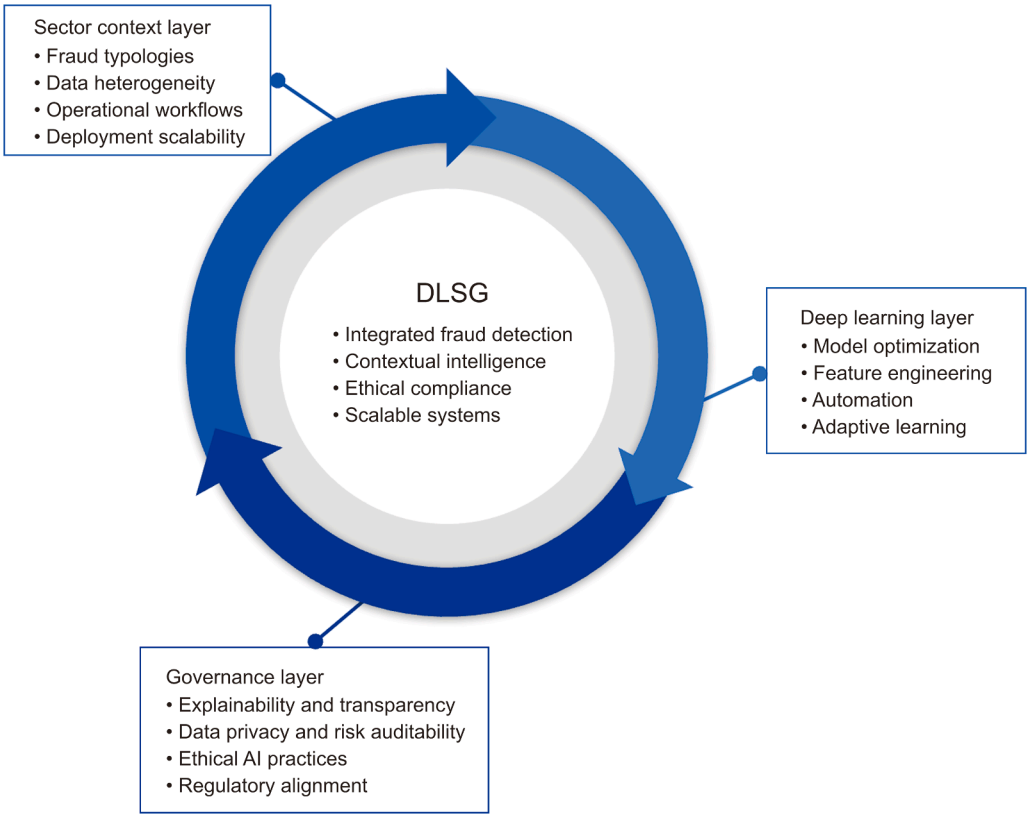


Fig. 10. The deep learning–sector–governance (DLSG) framework for building scalable and compliant financial fraud-detection systems.

fraud-detection, leading to measurable gains in accuracy and efficiency (Aljofey et al., 2022; Chen, 2020; Kaafarani et al., 2024). Meanwhile, banking and payment systems are confronted with increasingly global, high-volume fraud schemes such as money laundering, necessitating scalable graph-based models and integration of regulatory datasets such as Know Your Customer (KYC) profiles.

Each sector has distinct operational priorities. Banking and payment systems require the highest scalability and support millions of transactions daily across diverse platforms. Credit card systems manage the largest volume of structured transaction data and demand the fastest real-time processing, often making fraudulent decisions within milliseconds at the point of sale or during online checkouts. In comparison, the insurance and blockchain sectors prioritize privacy preservation, data integrity, and regulatory auditability over ultra-low latency processing. These sector-specific distinctions highlight the critical need for adaptive, context-aware fraud-detection architectures and reinforce the importance of sectoral alignment within the DLSG framework.

To address the concern of theoretical isolation, the DLSG framework was designed to complement and extend existing paradigms, such as the cross-industry standard process for data mining and findable, accessible, interoperable, reusable data principles, by embedding DL interpretability and governance constraints directly into the system architecture design.

The DLSG framework provides a practical roadmap for practitioners in finance, insurance, and fintech to deploy scalable, regulation-ready fraud detection systems. In industries governed by strict compliance standards, such as insurance and banking, risk and compliance officers must prioritize models that are not only accurate but also explainable and auditable. IT leaders can operationalize DLSG by adopting privacy-preserving architectures, such as federated learning and blockchain, to facilitate secure, compliant data collaboration. Strategic investments in real-time analytics infrastructure, informed by the DLSG framework, help organizations ensure scalability with rising transaction volumes and system complexity (Campanella et al., 2023).

Effective deployment also requires the selections of domain-specific models: transformer architectures for sequential transaction streams in banking,or federated learning for distributed insurance data. Robust system integration is an essential-model that must be embedded into real-time decision engines, deployed via modular application programming interfaces (APIs), and continuously improved through adaptive monitoring. At the governance level, incorporating XAI tools ensures transparency and auditability, alignment of systems with stakeholders, and legal expectations.

The third pillar of the framework is interwoven with these technical and sectoral dimensions: governance. Data privacy, anonymization, and regulatory compliance are not external constraints but are integral to fraud-detection design. Policies such as the GDPR, CCPA, and sector-specific mandates such as HIPAA fundamentally shape the permissible scope of data use and mandate interpretability in algorithmic decision-making (Krishnan et al., 2023). Techniques such as PCA, secure multi-party computation, and blockchain-based logging support the development of fraud-detection systems that are both operationally effective and legally compliant. Moreover, explainability is now a mandatory requirement in high-stakes sectors, such as banking and insurance, where model outputs may trigger financial penalties, affect claim approvals, or initiate regulatory investigations. Ensuring that deep learning models satisfy the “right to explanation” under GDPR requires the systematic integration of interpretability tools throughout the model development and deployment pipeline.

The DLSG framework conceptualizes these three components, namely DL, sector context, and governance, as mutually reinforcing layers. DL provides the adaptive intelligence necessary to detect the evolving fraud patterns. Sector-specific requirements guide the model architecture, evaluation criteria, and deployment strategies. Governance frameworks enforce transparency, privacy, fairness, and accountability throughout the system lifecycle. The novelty of the DLSG lies in its triadic synthesis—positioning itself as a meta-framework that bridges the gap between technical innovation and sectoral-compliance needs, an area underexplored in traditional fraud-detection literature. For practical applications, each dimension is mapped to real-world practices and technologies. For example, DL modules are categorized by fraud pattern type (e.g., sequential vs. topological), sectoral contexts are aligned with compliance and latency requirements, and governance is supported by automated transparency tools and standardized audit protocols. Fraud-detection systems can achieve both operational viability and societal trustworthiness only through the deliberate alignment of all three dimensions.

The DLSG framework identifies several key priorities for future research. There is a pressing need to develop standardized, sector-specific benchmarks that evaluate not only predictive accuracy but also economic impacts, such as fraud loss prevention and false positive costs. As federated learning and blockchain integration technologies mature, further research should explore their interoperability and capacity for real-time fraudulent adaptation. Additionally, hybrid and transfer learning models that span multiple financial sectors may offer promising avenues in environments with sparse or fragmented labeled data. Finally, explainable DL must remain a central focus, with continued innovation aimed at making fraud-detection models auditable, justifiable, and aligned with evolving stakeholder and regulatory expectations.

Ultimately, the DLSG framework not only synthesizes the current state of the field but also offers a practical blueprint for building secure, scalable, and regulation-ready fraud-detection infrastructures. It is designed to support decision-makers across financial institutions, regulatory bodies, and AI development teams in aligning technical innovation with legal compliance and sector-specific realities.

In summary, the DLSG framework offers a holistic blueprint for the development of the next-generation financial fraud-detection systems. By explicitly integrating deep learning capabilities with sector-specific operational demands and legal governance frameworks, the DLSG framework supports the creation of adaptable, secure, and trustworthy fraud mitigation solutions that can meet the evolving needs of the global financial ecosystem in both research and practice.

6. Future directions

Although this study reviews advancements in DL for financial fraud-detection, several limitations remain. First, the restriction to studies published between 2019 and 2024 excludes earlier foundational research that could provide historical context and evolutionary trends. Second, although prioritizing publicly available literature ensures transparency, it does not focus on proprietary models or industry-specific methodologies, which may offer valuable insights into real-world challenges. Additionally, heavy reliance on synthetic or publicly available datasets raises concerns about the operational robustness of many models, as real-world fraud cases often present far greater complexity. Finally, the exclusive focus on English-language publications restricts perspectives from regions where different fraud patterns and regulatory frameworks prevail, potentially overlooking important localized strategies.

Future research should prioritize standardization across the data science lifecycle to enhance reproducibility, comparability, and practical relevance. In particular, the development of unified data preprocessing pipelines, including techniques such as SMOTE, GAN-based augmentation, and feature transformation, would help address the pervasive class imbalance and ensure greater consistency across studies. Establishing globally recognized benchmark datasets that capture diverse fraud typologies is critical for a fair model evaluation and cross-sector applicability.

Advances in model development must continue, particularly through collaboration between academia, industry, and regulatory agencies. Hybrid approaches that integrate supervised, unsupervised, and reinforcement learning offer promising pathways for building adaptive

fraud-detection systems. Moreover, the incorporation of XAI techniques and automated hyperparameter tuning should be prioritized to enhance both interpretability and scalability (Černevičienė and Kabašinskas, 2024).

Comprehensive model evaluation frameworks need to be extended beyond traditional classification metrics. Future studies should adopt evaluation criteria that reflect the economic impact, cost of false positives, and feasibility ofoperational deployment. The creation of public leaderboard systems, in which models are continuously benchmarked against evolving fraud scenarios under real-world constraints, would drive more rigorous model innovation.

Research must also address the seamless integration of DL models into existing financial infrastructures. The exploration of robust API-based deployment strategies and modular architectures can facilitate operational adoption. In parallel, privacy-preserving techniques such as federated learning and homomorphic encryption, should be embedded into fraud-detection models to ensure compliance with regulations such as GDPR and CCPA.

To translate research into real-world utility, institutions should develop pipelines that integrate detection models with core transaction systems, design modular architectures for sector-specific use, and implement continuous monitoring of concept drift. These actions will help align technological solutions with compliance, cost-efficiency, and scalability goals.

Finally, there is an urgent need for adaptive model monitoring and maintenance strategies. Self-learning systems that are capable of detecting concept drift, meta-learning frameworks for continuous model updating, and blockchain-based fraud monitoring mechanisms (Sharma et al., 2024) represent promising avenues. These approaches would ensure that fraud-detection systems remain resilient and effective as fraud patterns evolve.

By focusing on these research directions, future studies can contribute to the development of scalable, interpretable, and regulation-compliant DL applications, strengthening the next generation of fraud prevention frameworks in the digital financial ecosystem.

7. Conclusion

This study highlights the pivotal role of DL in reshaping financial fraud-detection within the rapidly evolving digital landscape. Through a comprehensive review of research published between 2019 and 2024, it is clear that DL models, including CNNs, LSTM networks, transformers, and ensemble architectures, have significantly advanced the detection of complex and dynamic fraud patterns across diverse financial sectors. Coupled with sophisticated preprocessing strategies and feature engineering techniques, these models have addressed longstanding challenges such as class imbalance, data sparsity, and the need for scalable solutions capable of real-time detection.

Equally important is the growing integration of privacy-preserving mechanisms and regulatory compliance frameworks into fraud-detection systems. Techniques such as PCA, federated learning, and blockchain-based architectures facilitate the secure and ethical use of sensitive data while aligning with global regulations such as the GDPR and the CCPA. These developments not only enhance individual data protection but also build institutional trust, paving the way for the broader adoption of advanced fraud mitigation strategies. Moreover, the increasing use of automation tools, including parameter tuning, subsampling, and end-to-end pipelines, has further accelerated fraud-detection processes, improving the operational efficiency in high-risk, high-volume environments.

By introducing the DLSG framework, this study provides a structured foundation for future innovations in financial fraud prevention. The DLSG framework emphasizes the critical alignment between algorithmic advancement, sector-specific operational requirements, and data governance considerations, reinforcing the necessity for fraud-detection systems that are not only technically robust but also operationally viable and compliant with regulatory standards. Researchers and practitioners must continue investing in interpretable and adaptive models, exploring emerging fraud vectors, and fostering collaborative ecosystems that enable secure, cross-institutional data sharing.

Ultimately, the insights from this review offer a roadmap for developing resilient, transparent, and trusted fraud detection solutions. By thoughtfully integrating cutting-edge ML techniques with practical operational strategies and governance frameworks, the financial industry can advance toward a more secure, accountable, and future-ready digital economy.

CRediT authorship contribution statement

Yisong Chen: Writing – original draft, Project administration, Methodology, Formal analysis, Conceptualization. **Chuqing Zhao:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Yixin Xu:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Data curation. **Chuanhao Nie:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Yixin Zhang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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